

Management of Elevated Lactate

Outline

Differential diagnoses

Investigations

Management

Dichloroacetate protocol

Arginine protocol for MELAS patients

Normal lactate values:	Blood < 2.1 mmol/L CSF < 2.2 mmol/L
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Sample must be obtained from vein (without cuffing/tourniquet) or artery; patient must be relaxed. Obtain specimen with a Na-fluoride sample tube (grey top).

DIFFERENTIAL DIAGNOSES

It is often difficult to distinguish primary and secondary causes of lactic acidemias. In addition it is important to note that a normal lactate level does not exclude a mitochondrial disorder.

Secondary causes:

- Use of tourniquet or difficulty in drawing blood (most common)
- Muscular activity, assisted ventilation, seizures (lactate up to 4 – 6 mmol/L)
- Severe systemic disease e.g. central or peripheral hypoxia/ischemia, shock, cardiac failure, cardiomyopathy, liver or renal failure, septicemia, diabetes mellitus, etc.
- Any severe metabolic disease
- Renal tubular syndrome, hyperchloremia, UTI
- Drugs (biguanides), alcohol intoxication
- Thiamine deficiency

Metabolic causes:

- Disorders of the respiratory chain or tricarboxylic acid cycle
- Pyruvate dehydrogenase (PDH) or pyruvate carboxylase deficiency
- Long chain fatty acid oxidation disorders
- Organic acidurias, disorders of biotin metabolism
- Glycogen storage diseases, gluconeogenesis disorders

INVESTIGATIONS

1. Arterial blood gas, urea and electrolytes and calculate the anion gap.
2. Serial lactate levels. A lactate level > 5 mmol/L requires urgent management with dichloroacetate and sodium bicarbonate. Consider doing CSF lactate if there are neurological deficits or impairment of consciousness.
3. Creatine kinase
4. Ammonia (send ammonia to lab immediately on ice)
5. CSF protein may be high in Kearns Sayre syndrome. Measure CSF amino acids as necessary.
6. Acylcarnitine analysis for fatty acid oxidation disorders
7. Plasma amino acids – elevated alanine reflects the concentration of pyruvate and may be seen in primary lactic acidemias
8. Urine organic acids
9. Ketosis is indicative of a primary metabolic disorder (inhibition of the tricarboxylic acid cycle and is not found in PDH deficiency or fatty acid oxidation disorders)
10. Postprandial elevation of lactate (>20%) or ketone bodies (paradoxical ketonemia) may indicate PDH deficiency or respiratory chain defects. Increased lactate after glucose challenge also in glycogen storage diseases Type O, III, and VI.
11. Postprandial fall of lactate and fasting hypoglycemia may indicate glycogen storage disease type I or gluconeogenic defects.
12. MRI brain, consider MR spectroscopy
 - Hyperintense signals in putamen, globus pallidus, caudate nucleus, brainstem
 - Stroke-like picture (not confined to vascular territory)
 - Leukodystrophy
 - Cerebral or cerebellar atrophy
 - High lactate peak on MRS of brain.
13. Muscle biopsy
 - Light microscopy (to include modified Gomori trichrome stain, COX, NADH and SDH staining)
 - Electron microscopy (need glutaraldehyde tube)
 - Snap frozen muscle in liquid nitrogen – for enzyme studies

MANAGEMENT

Consult a metabolic physician.

General Treatment Measures:

1. Ensure adequate intake of energy, fluids and electrolytes.
2. Restrict glucose intake (if on parenteral nutrition start with 2 – 4 mg/kg/min and monitor hypocounts), add lipids (1 – 2 g/kg/day) when fatty acid oxidation

disorders have been excluded. Fat component of diet should be 50 – 60% daily caloric intake

3. Avoid/treat conditions that have high energy consumption
 - Treat fever efficiently
 - Treat seizures efficiently, avoid valproic acid
4. **Avoid** drugs that may inhibit the respiratory chain
 - Aminoglycoside antibiotics
 - Beta blockers
 - Chloramphenicol
 - HMG-CoA reductase inhibitors (statins)
 - Ringer's lactate solution
 - Metformin
 - Neuromuscular blocking agents
 - Nucleoside reverse transcriptase inhibitors
 - Phenobarbital
 - Propofol
 - Serotonin receptor agonists (triptans)
 - Tetracycline
 - Theophylline
 - Valproic acid

Treat Acidosis

1. Consider using a buffer if pH is <7.2
2. Use sodium bicarbonate. Check ABG as necessary.
3. Use THAM buffer if sodium is high.
4. Use dichloroacetate if lactate is > 5 mmol/L (protocol for dichloroacetate use below). Dichloroacetate use should be minimized because of risks of peripheral neuropathy.
5. Consider dialysis if lactate is severe. Avoid use of lactate-containing dialysate.

Use of vitamins and cofactors:

1. **Coenzyme Q10 4 – 10 mg/kg/day**
2. Biotin 0.5mg/kg/day (max 200mg/day – not per kg)
3. **Carnitine 50 – 100 mg/kg/day in 2 – 3 divided doses (maximum 3 g/day) (after excluding fatty acid oxidation disorder; check carnitine levels)**
4. Vitamin C 5 - 10 mg/kg/day in 1 – 3 divided doses
5. Vitamin E (α-tocopherol) 1 – 2 IU/kg/day in 1 – 3 divided doses
6. **Vitamin B1 (Thiamine) 1 – 2 mg/kg/day**
7. **Vitamin B2 (Riboflavin) 50 – 100 mg/day (not per kg)**
8. Vitamin B3 (Nicotinamide) 50 mg/day (not per kg)
9. Folinic acid 3 – 5 mg/kg/day

DICHLOROACETATE PROTOCOL

DCA maintains pyruvate dehydrogenase in its active form, encouraging the production of pyruvate and the decrease in lactate.

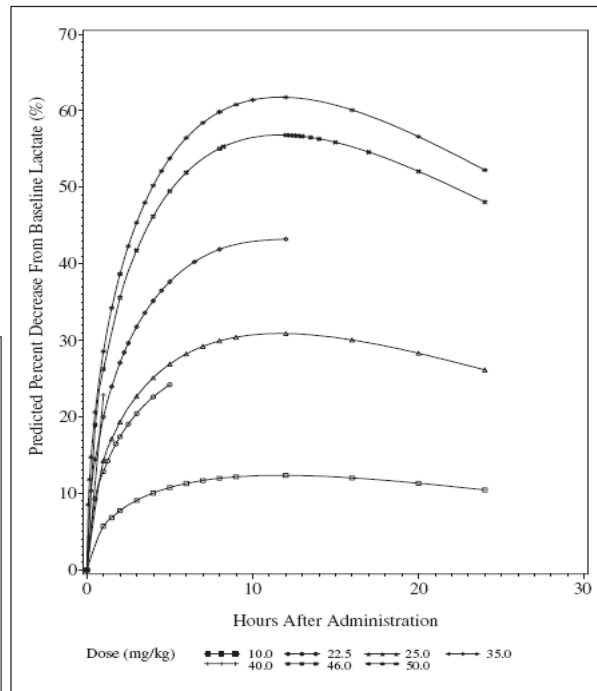
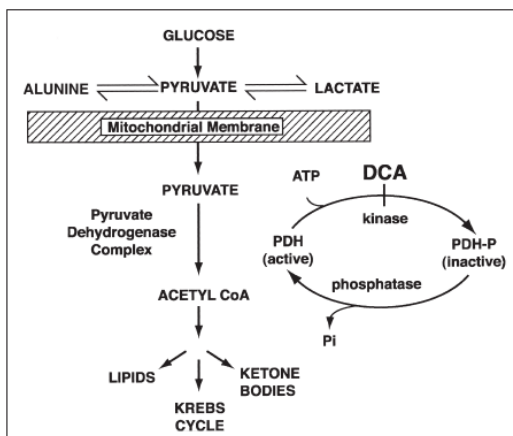


Figure 4. Percent decrease from baseline lactate (mmol/L) over time following a single administration of dichloroacetate (DCA) as a function of the dose levels contained in the meta-analysis data set.

(from Stacpoole J Clin Pharmacol 2003)

Bicarbonate therapy is the mainstay of correction of acidosis. However, this may only transiently raise plasma bicarbonate concentration. It can also lead to a variety of problems such as fluid overload, hypernatremia and post-recovery metabolic alkalosis. Nonetheless, sodium bicarbonate is indicated for acute control of acidemia. Alternative buffering agents such as THAM and DCA may also help to decrease lactate production.

1. Check lactate level. If > 5.0 mmol/L and suspicion is high for a **primary** lactic acidosis, consider the use of dichloroacetate.
2. IV Dichloroacetate is available in the PICU (x 5459, 3409)
3. **Administer IV dichloroacetate 100 mg/kg/dose for infants and children (to be infused over 1 hour) every 12 hours as necessary.**
Dose for adolescents and adults 25-50mg/kg/dose
4. Check lactate level at baseline (0 hours), 4 hours, 8 hours and 12hours. Repeat the DCA dose only if lactate is > 5.0 mmol/L. The figure above shows that from meta-analysis of various studies, there is a predicted decrease of about 10 – 60% in lactate level about 10 – 12 hours after administration of DCA.

ARGININE PROTOCOL FOR MELAS PATIENTS

Arginine use in MELAS encourages vasodilation through nitric oxide synthesis and decreases incidence of stroke-like episodes as well as the severity of strokes when administered early in the event.

*In the event of any neurological deficit noted in a MELAS patient, IV Arginine must be administered within **4 – 6 hours**.*

ACUTE DOSE OF ARGININE

IV Arginine 21% solution (4.2g in 20mL injection)

- **Compatible with D5% and D10% solutions (No data with Normal Saline)**
- **Recommended dilution to 20mg/mL (max 100mg/mL in Central Line)**

Dosing = 0.5 g/kg/dose over 30 minutes (children)

Dosing = 10g/m² body surface area (adolescents and adults)

$$BSA(m^2) = \sqrt{\frac{Height(cm) \times Weight(kg)}{3600}}$$

Infusion to be repeated every 6 hours for the next 24 – 48 hours depending on the resolution of symptoms. *Infusion should not be repeated more than for 24 hours if the patient is already on maintenance arginine previously or is known to have a good arginine level.*

MAINTENANCE DOSE OF ARGININE

Oral L-Arginine 0.15 – 0.3 g/kg/day in 2 – 3 divided doses.

In refractory disease, consider higher dose of 0.3 – 0.5 mg/kg/day in 4 divided doses

MONITORING:

STAT on admission:

- ABG, Full blood count, glucose, renal panel #2, ammonia, lactate
- Plasma amino acids (before arginine infusion)

1 hour after each IV arginine dose: ABG and glucose

- **Lactate levels 4 – 6 hourly in acute period**

Outpatient surveillance:

- Lactate level as necessary
- Plasma amino acids every 4 – 6 months (or more frequently if strokes recur at shorter intervals).

Monitor adverse effects:

- IV formulation - hypotension, headaches, metabolic acidosis
- Oral formulation - nausea, vomiting

REFERENCES

1. Margaret O James and Peter W Stacpoole. Pharmacogenetic considerations with dichloroacetate dosing. Pharmacogenomics 2016 May; 17(7):743-753
2. <http://ageps.aphp.fr/wp-content/blogs.dir/67/files/2019/10/ESCP18SY-1064-Arginine-et-Citrulline.pdf>
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